

16/17 Sem 2 Q3 (b)

Show that

$$|\underline{u} \times \underline{v}|^2 = |\underline{u}|^2 |\underline{v}|^2 - (\underline{u} \cdot \underline{v})^2,$$

$\underline{u}, \underline{v}$  are vectors in  $\mathbb{R}^3$ .

Sol.

Observation:

$$|\underline{u} \times \underline{v}| = |\underline{u}| |\underline{v}| |\sin \theta|$$

$$|\underline{u} \cdot \underline{v}| = |\underline{u}| |\underline{v}| |\cos \theta|$$

$$|\underline{u} \times \underline{v}|^2 = |\underline{u}|^2 |\underline{v}|^2 |\sin \theta|^2$$

$$= |\underline{u}|^2 |\underline{v}|^2 \sin^2 \theta$$

$$= |\underline{u}|^2 |\underline{v}|^2 (1 - \cos^2 \theta)$$

$$= |\underline{u}|^2 |\underline{v}|^2 - |\underline{u}|^2 |\underline{v}|^2 \cos^2 \theta$$

$$= |\underline{u}|^2 |\underline{v}|^2 - (\underline{u} \cdot \underline{v})^2$$

$$= |\underline{u}|^2 |\underline{v}|^2 - (\underline{u} \cdot \underline{v})^2$$